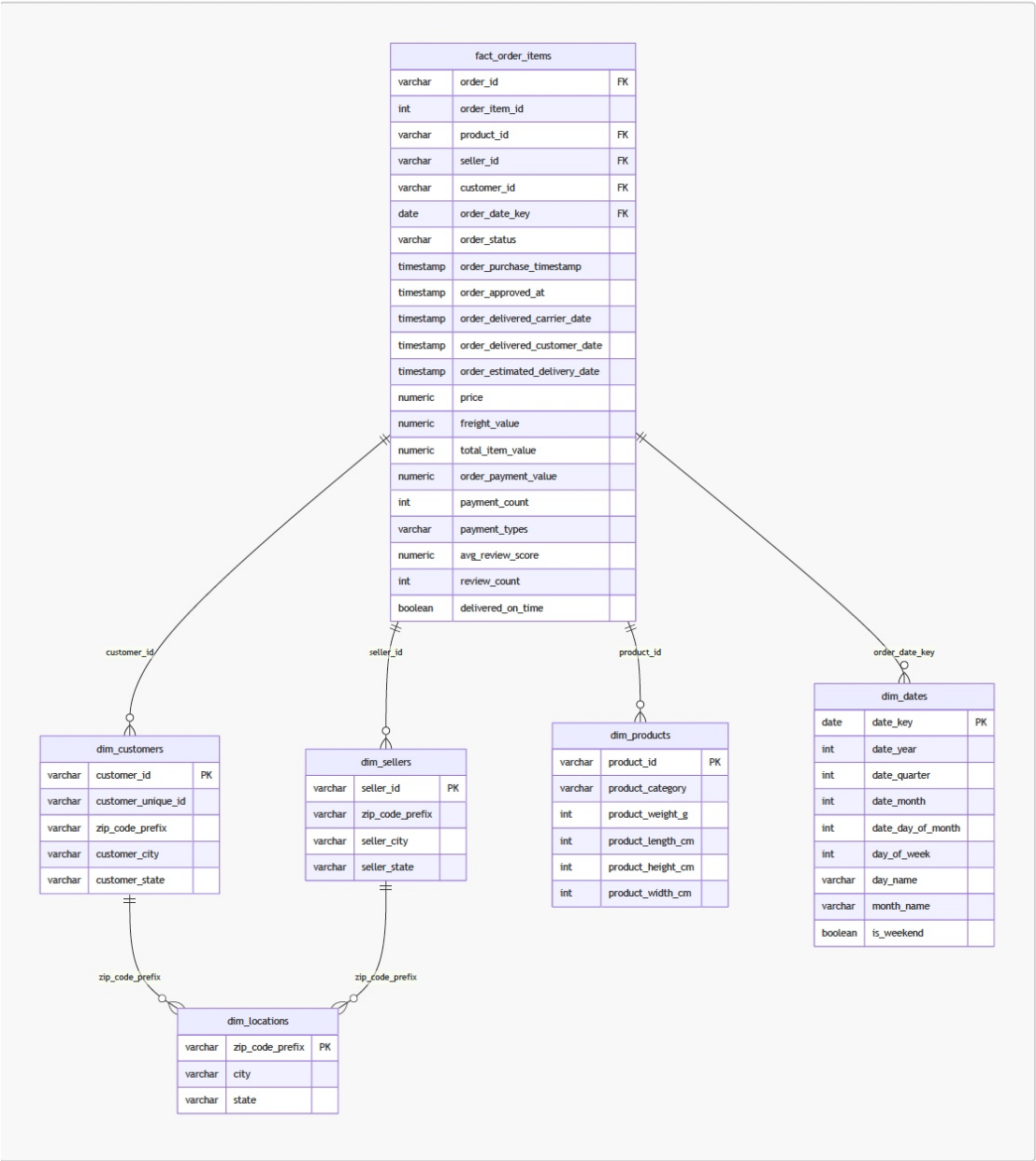
Olist Star Schema Diagram

Overview

This star schema transforms the Olist OLTP database (3NF) into an optimized analytical model with one fact table and five dimension tables.



Design Decisions

Aspect	**Decision**	**Rationale**
Grain	One row per order line item	Most granular level; supports any aggregation
Fact Type	Transaction fact table	Each row represents a business event (item sold)
Date Dimension	Generated from order timestamps	Covers only dates with actual orders
Location	Shared dimension for customers & sellers	Avoids duplication of geographic data
Payments	Aggregated to order level in fact	Payments are per-order, not per-item
Reviews	Averaged to order level in fact	Reviews are per-order, not per-item
Materialization	Dimensions and fact as tables; staging as views	Tables for query performance; views for freshness